ClipSAT : Egyptian Scholastic Test: EST II Mathematics Level 2

Unit 6.01: Permutations, Combinations & Conditional Probability Name: _____

Q1. If i = sqrt{-1} , which of the following is equivalent to the complex expression frac{7 + 4i}{2 - i} ?

[ ] A: 2 + 3i
[ ] B: 2 - 3i
[ ] C: frac{10 + 15i}{3}
[ ] D: frac{18 + 15i}{5}

Q2. The graph of the function y = fx is shown below. What is the value of f1 ?

[ ] A: 0
[ ] B: 2
[ ] C: -1
[ ] D: 3

Q3. Consider the function fx = x^2 - 4x + 3 . What are the x-intercepts of the graph of this function?

[ ] A: 1, 0 and 3, 0
[ ] B: 0, 1 and 0, 3
[ ] C: -1, 0 and -3, 0
[ ] D: 1, 0 and -3, 0

Q4. A vector mathbf{v} is represented by the directed line segment from point P1, 2 to point Q4, 6 . What are the components of vector mathbf{v} ?

[ ] A: [3, 4]
[ ] B: [5, 8]
[ ] C: [-3, -4]
[ ] D: [4, 3]

Q5. A bag contains 12 red marbles and 8 blue marbles. If one marble is drawn at random, what is the probability that the marble is red?

[ ] A: frac{3}{5}
[ ] B: frac{2}{5}
[ ] C: frac{1}{2}
[ ] D: frac{3}{4}

Q6. Consider an arithmetic sequence where the first term is a_1 = 5 and the common difference is d = 3 . What is the value of the 4th term, a_4 ?

[ ] A: 14
[ ] B: 17
[ ] C: 8
[ ] D: 11

Q7. A geometric sequence has a_1=6 and common ratio r=2 . Find a_{5} .

[ ] A: 96
[ ] B: 192
[ ] C: 98
[ ] D: -96

Q8. The graph of the function y = fx = ax^2 + bx + c is shown below. The graph passes through the points -1, 5 , 1, 1 , and 3, 5 . Determine the value of b .

[ ] A: 0

[ ] B: -2

[ ] C: 2

[ ] D: 1